

Name - Mainpal

Subject - Java Programming

Assignment - 3

Roll No - 2401730097

Assignment-3

0 - ?

```
# import java.util.*;
```

```
class InvalidMarksException extends Exception {
    public InvalidMarksException (String message) {
        super (message);
    }
}
```

```
class student {
    int rollNumber;
    String studentName;
    int[] marks;
```

```
Student (int rollNumber, String studentName, int[] marks) {
    this.rollNumber = rollNumber;
    this.studentName = studentName;
    this.marks = marks;
}
```

```
void valid validateMarks () throws InvalidMarksException {
    for (int m: marks) {
        if (m < 0 || m > 100) {
            throw new InvalidMarksException ("Marks must be
            b/w 0 and 100");
        }
    }
}
```



```
double calculate Average () {
    int sum = 0;
    for (int m: marks) sum += m;
    return sum / 3.0;
}
```

```
void display Result () {
    System.out.println("Roll Number:" + rollNumber);
    System.out.println("Name:" + studentName);
    System.out.println("Marks:" + Arrays.toString(marks));
    double avg = calculate Average ();
    System.out.println("Average:" + avg);
}
```

```
    boolean pass = true;
    for (int m: marks) {
        if (m < 35) pass = false;
    }
    System.out.println("Result:" + (pass ? "pass" : "Fail"));
}
```

```
public class Main {
    Student[] students = new Student[100];
    int count = 0;
    Scanner sc = new Scanner(System.in);
```

```
    void add Student () {
```

```
        try {
            System.out.print("Enter Roll Number:");
            int roll = sc.nextInt();
            sc.nextLine();
```



```
System.out.println("Enter Name:");
String name = sc.nextLine();
```

```
int[] marks = new int[3];
```

```
System.out.println("Enter 3 subject marks :");
```

```
for (int i = 0; i < 3; i++) {
    marks[i] = sc.nextInt();
}
```

```
Student s = new Student(roll, name, marks);
s.validate Marks();
```

```
Students[count++] = s;
```

```
System.out.println("Student Added Successfully");
```

```
} catch (InvalidMarksException e) {
```

```
System.out.println("Error : " + e.getMessage());
```

```
} catch (InputMismatchException e) {
```

```
System.out.println("Invalid input type");
sc.nextLine();
}
```

```
void showStudentDetails() {
```

```
System.out.println("Enter Roll Number:");
```

```
int roll = sc.nextInt();
```

```
for (int i = 0; i < count; i++) {
```

```
if (Students[i].rollNumber == roll) {
```

```
Students[i].display Result();
```

```
return;
```



```
System.out.println("Student not found");
}
```

```
void main Menu() {
```

```
    while (true) {
```

```
        System.out.println("1. Add Student");
```

```
        System.out.println("2. Show Student Details");
```

```
        System.out.println("3. Exit");
```

```
        System.out.println("Enter Choice:");
```

```
        int ch;
```

```
        try {
```

```
            ch = sc.nextInt();
```

```
        }
```

```
        catch (InputMismatchException e) {
```

```
            System.out.println("Invalid Choice!");
```

```
            sc.nextLine();
```

```
            continue;
```

```
        }
```

```
        if (ch == 1) addStudent();
```

```
        else if (ch == 2) showStudentDetails();
```

```
        else if (ch == 3) break;
```

```
        else System.out.println("Invalid choice");
```

```
    }
```

```
    System.out.println("Closing the program...");
```

```
    sc.close();
```

```
}
```

```
public static void main (String[] args) {
```

```
    Main m = new Main();
```

```
    m.main Menu();
```

```
}
```